Software Architecture II

Arquitectura de Software II

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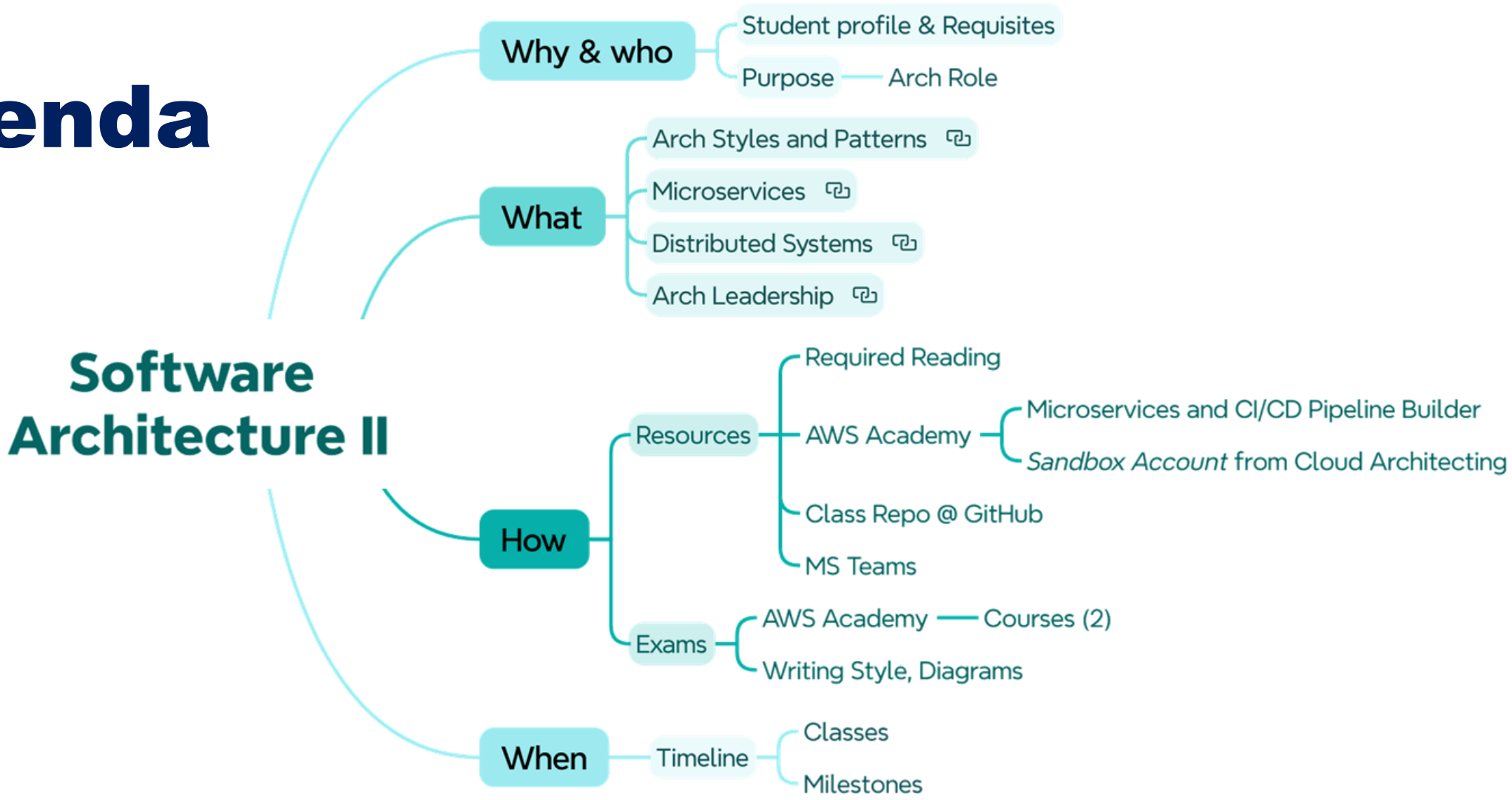
Agenda



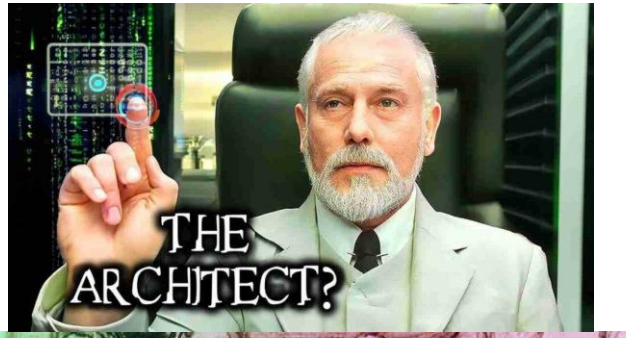
Why: Student Profile

- Curious and Discipline: Read IT topics on several sources and plan his/her works.
- Dev: AWS Labs use Node, but knowledge of a modern language is enough (Go, Rust, Clojure, Java, Python).
- Usage of GenAI Tools for specific tasks only.

Prerequisites:

- Containerization: Usage of Docker CLI. Course on this MSc (Arquitectura de Software I).
- Database: Knowledge on NoSQL DB Design. Course on this MSc (Diseño y Optimización de Bases de Datos).
- Cloud: Knowledge on AWS. Course on this MSc (Arquitecturas en la Nube). Focus on ECS, CodePipeline and CodeDeploy (Asynch Class).

Why: Architecture Role



- To set responsibilities of an architect role.
- To apply proven (industry) methods to solve problems.
- To learn how to make good things (best practices), possible things (constraints) and control exceptions (antipatterns).

See laws and expectations after R*equired Reading*

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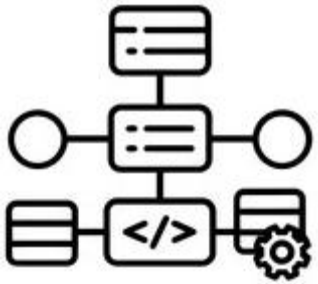
What

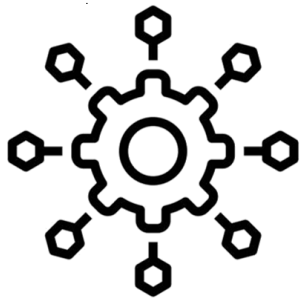
Estilos y Patrones Arquitectonicos

Architectural Styles and Patterns



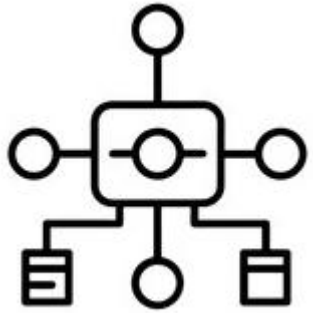
Microservicios

Microservices



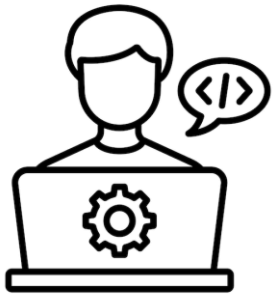
Sistemas Distribuidos

Distributed Systems
{Platforms, Infrastructure, Computing}



Liderazgo en Arquitectura

Architecture Leadership



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How: Resources

01 **AWS Academy**

02 **Class Repo@GitHub**

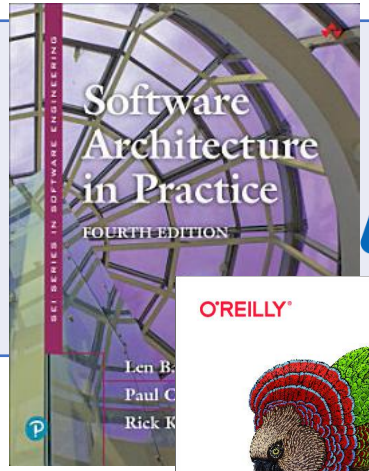
03 **MS Teams**

04 **Unisabana e-Learning**

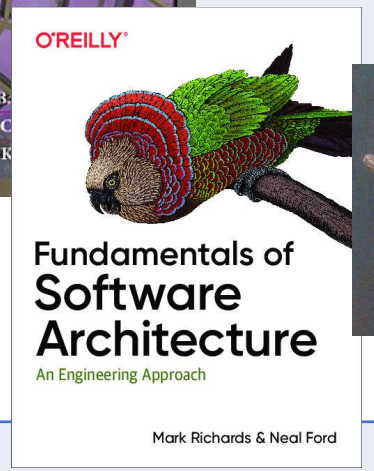
Note: Take advantage of AWS Academy promos. All links or books are referenced on the slide or at the end of presentation.

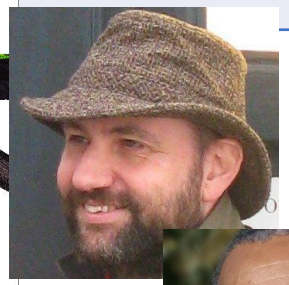
How: Required Reading

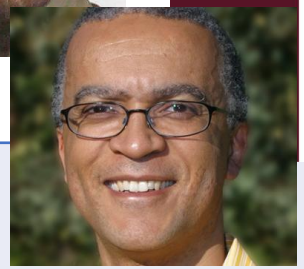
Architectural Styles and Patterns

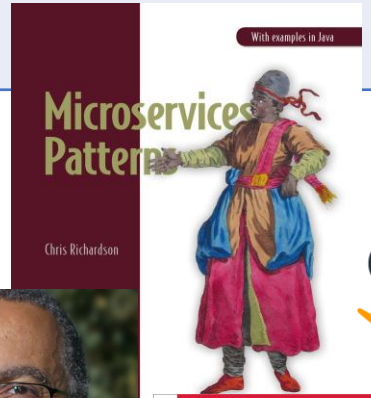


Microservices





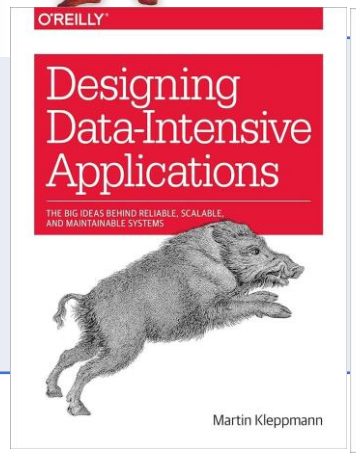


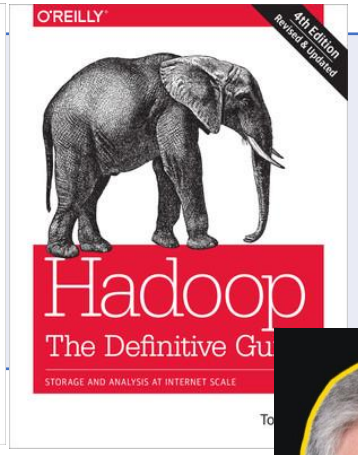


aws

Distributed Systems
{Platforms, Infrastructure, Computing}

Base





Architecture Leadership

[Many……]

See How my reference system at the end

How: Exams

Item	%	Deadlines	Type	On
Arch Styles and Patterns	20	3rd Class	Doc	Email/Uni e-learning
Microservices	50	5th Class	Doc & Lab	AWS Academy & Uni e-Learning
Distributed Systems	15	Last Class	Doc & Lab	AWS Academy & Uni e-Learning
Arch Leadership	15	Last Class	Docs (2)	Email/Uni e-learning

Proposals to run scripts for labs: Mondays, Thursdays or Fridays.
For anyday, the professor will run the scripts at 5 pm, if you send the credentials and specify the day.

AWS Academy issues: Create support the ticket directly and keep inform the professor. AWS Academy will need all screenshots.

Unisabana e-Learning issues: Review with University Point-of-Contact, the IT support.

Extra Time with regular process **only**: Review with University Point-of-Contact, the procedure and evidences.

Take care of the end date for Courses on AWS Academy to take advantage of the promos and labs.

AWS Academy

- Task 2: For this course, Microservices and CI/CD Pipeline Builder.
- Task 3: Take advantage of the Sandbox Account from previous course, “Cloud Architecting”.

Those tasks are evaluated by the professor using scripts. AWS Academy doesn’t offer a grade for those elements.

Preferred style

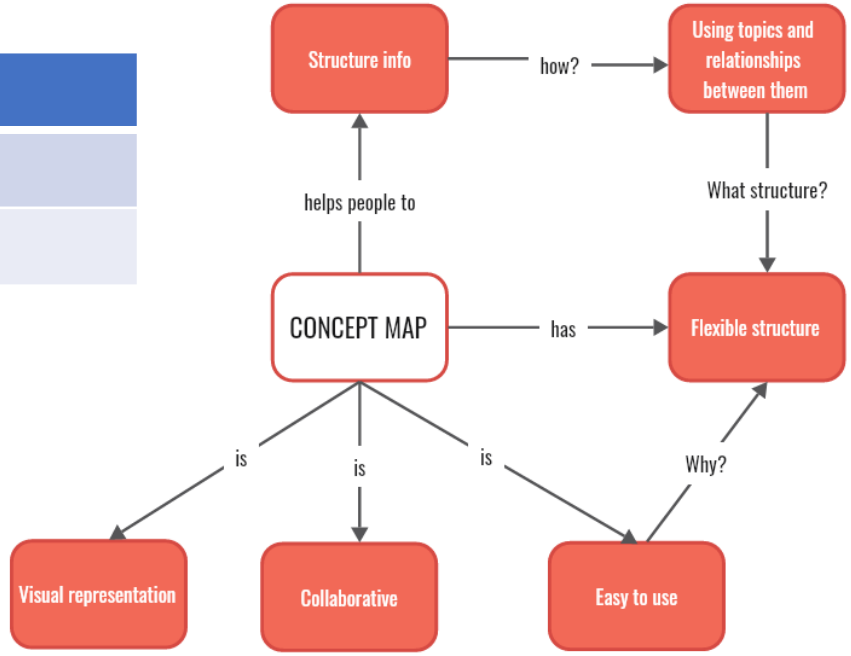
Simple

Well-Structured

Relevant info

Header1	Header2
Or using	A table
With clever	order

- Using bullets,
- It can be a
- Great idea!

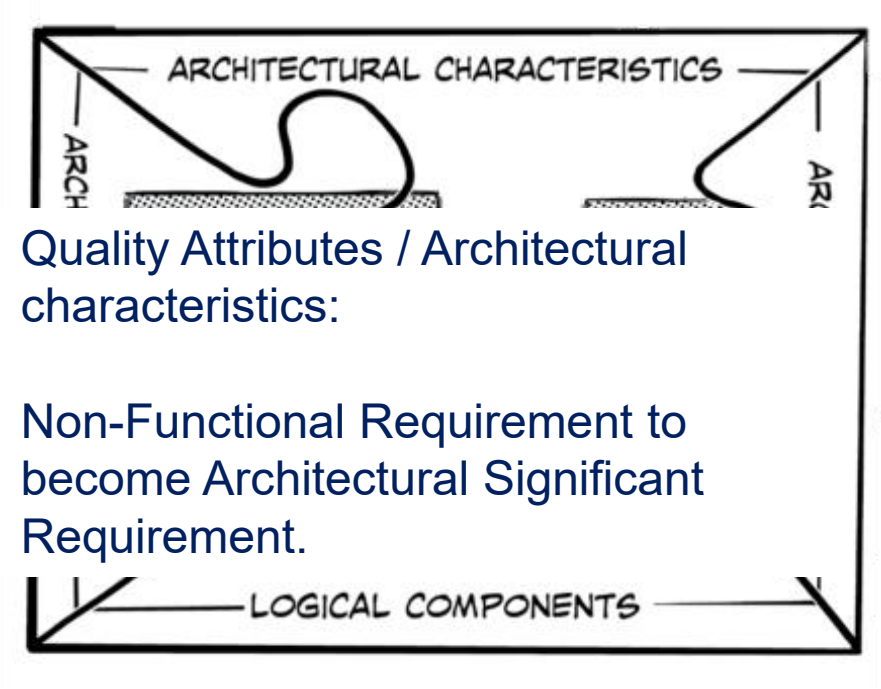


When: Timeline

* Review updated dates on Repo@GitHub.

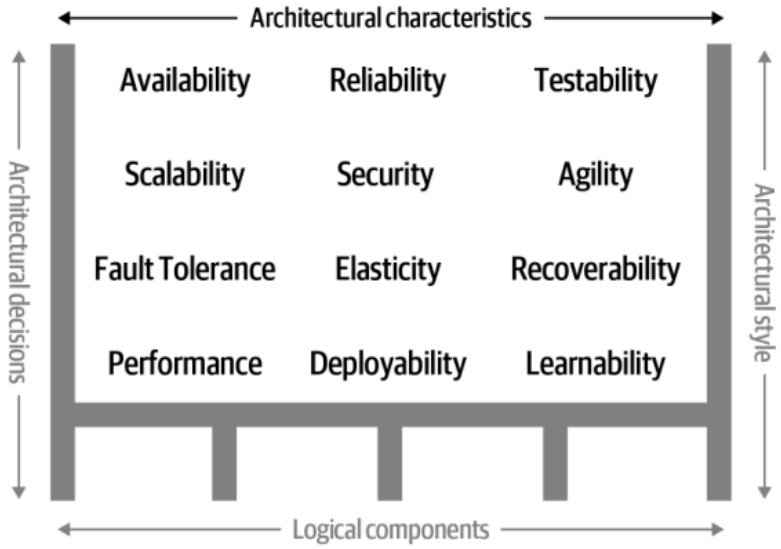
Class #	Date	Module	Milestone
1	- Jul	0, 1	
2	- Jul	1	
3	- Jul	2	Doc for Module 1
4	- Aug	2	
5	- Aug	3	Doc and lab for Module 2
6	- Aug	4	Doc and lab for Module 3, Doc for Module 4

SA Dimensions: Quality Attributes

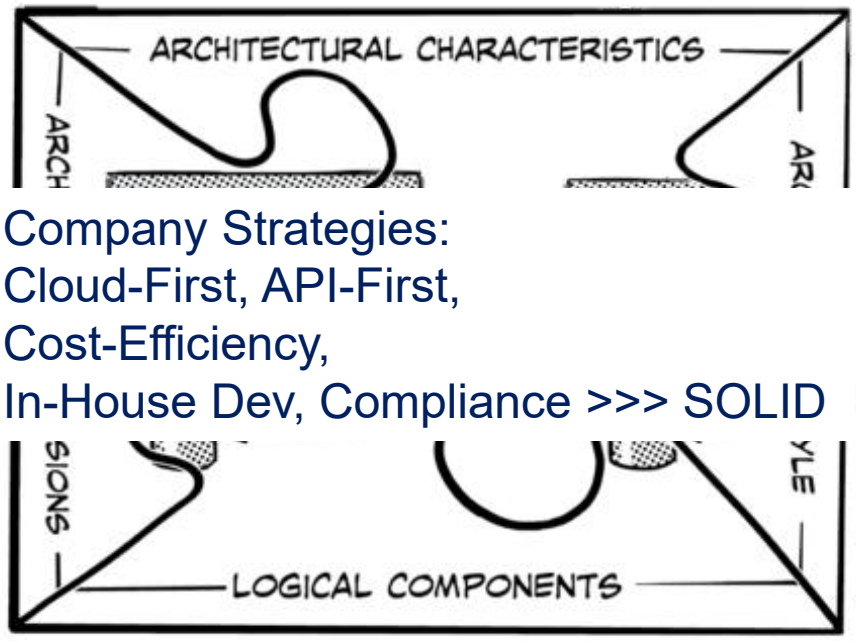


Quality Attributes / Architectural characteristics:

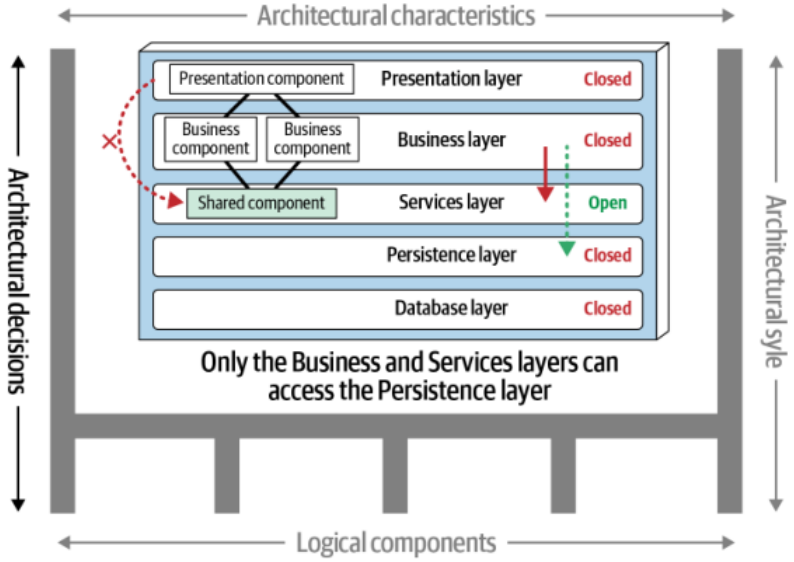
Non-Functional Requirement to become Architectural Significant Requirement.



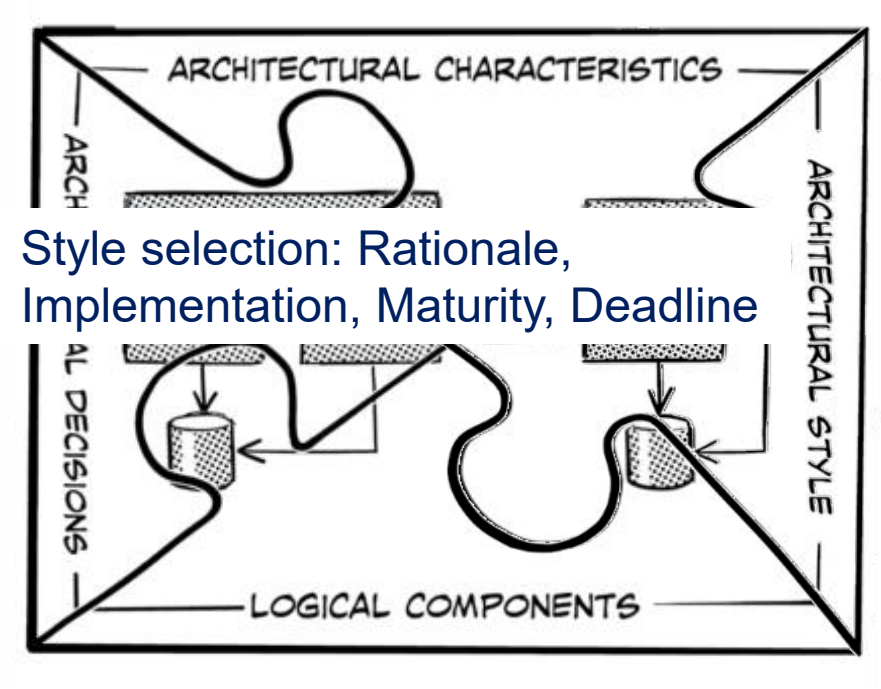
SA Dimensions: Architectural Decisions



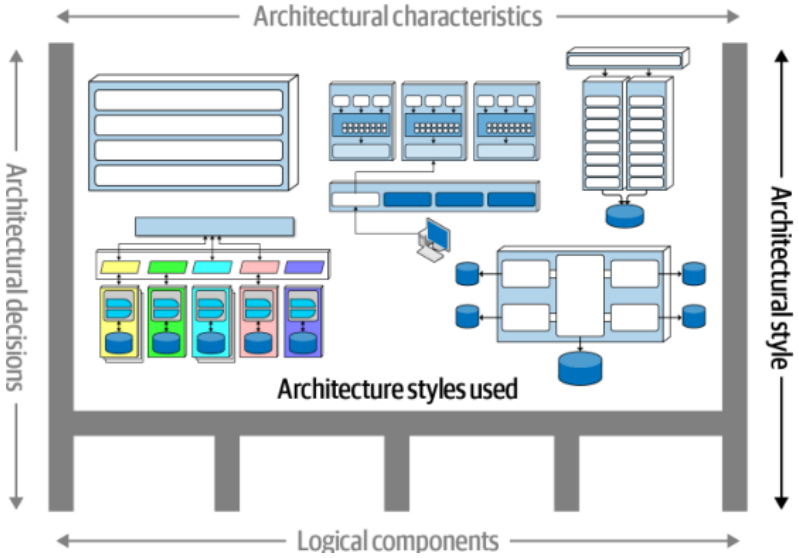
Company Strategies:
Cloud-First, API-First,
Cost-Efficiency,
In-House Dev, Compliance >>> SOLID



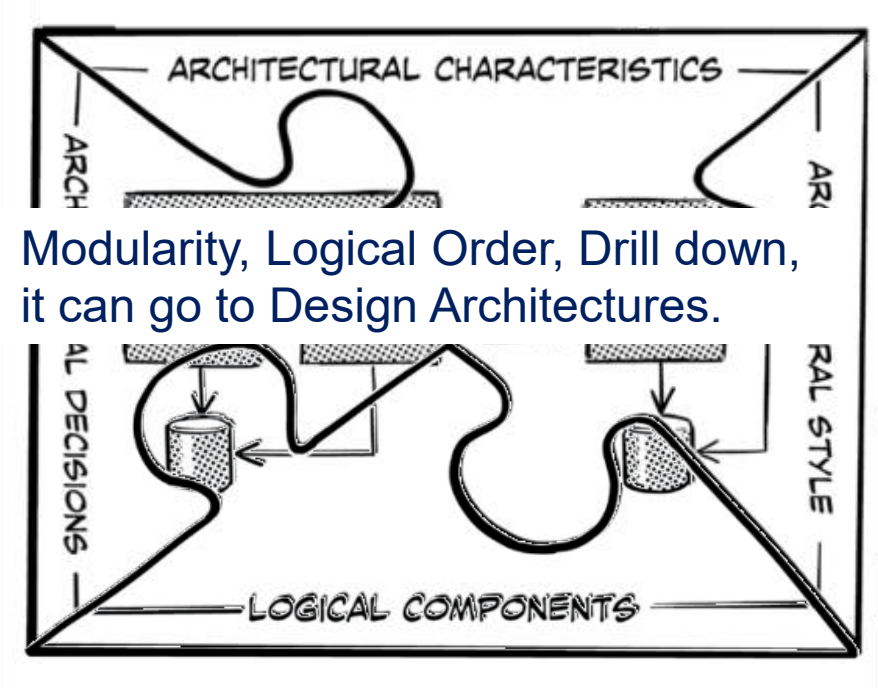
SA Dimensions: Architectural Styles



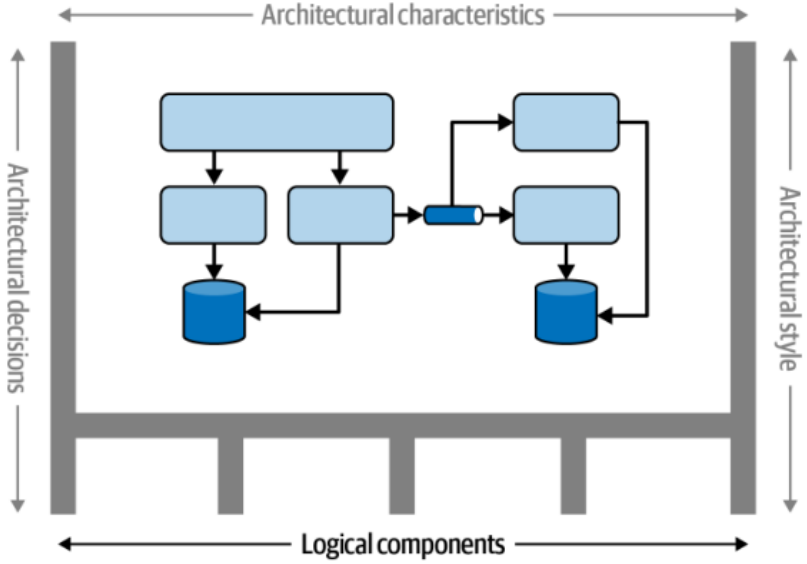
Style selection: Rationale, Implementation, Maturity, Deadline



SA Dimensions: Logical Components



Modularity, Logical Order, Drill down, it can go to Design Architectures.



Scope and Laws

Leadership

- Transparent
- Discipline
- Original

Laws of Software Architecture

1. Everything in software architecture is a trade-off.
2. Why is more important than how.
3. Most architecture decisions aren´t binary but rather exist on a spectrum between extremes.

Expectations

- Make architecture decisions
- Continually analyze the architecture
- Ensure compliance with decisions
- Keep current with latest trends
- Understand diverse technologies, frameworks, platforms, and environments
- Know the business domain
- Lead a team and possess interpersonal skills
- Understand and navigate organizational politics

Good Quotes

Architecture is the stuff you can’t Google or ask an LLM about.
—Mark Richards

There are no right or wrong answers in architecture—only trade-offs.
—Neal Ford

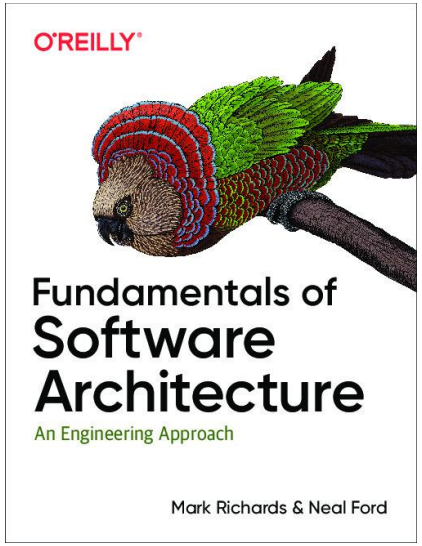
Programmers know the benefits of everything and the trade-offs of nothing.
Architects need to understand both.
—Rich Hickey, Clojure Creator

References

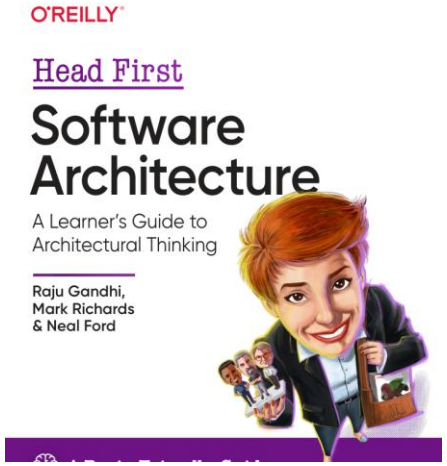
Links are everywhere can be exasperating, so it is better to have an ending slide with the chapters and the acronym.

The abbreviations to be referred are here.

- [FoSA] Richards, M., & Ford, N. (2025). Fundamentals of software architecture (2nd ed.). O'Reilly Media.
  Ch. 1, 26 & 27



- [HFSoA] Raju, S., Richards, M., & Ford, N. . (2024). Head First software architecture: A learner's guide to architectural thinking (1.ª ed.). O'Reilly Media.
  Ch. 1, 2 & 3



[FoSA] and [HFSoA]

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